

Optimization 1

ISE 5406

Lecture 4

Dr. Robert Hildebrand

Grado Department of Industrial & Systems Engineering
Virginia Tech

Where We Are in the Course

Last Time (Lecture 3): Real Analysis and Optimization

- Closed, open, compact, and bounded sets
- Weierstrass' Theorem for existence of optimal solutions
- Separating and supporting hyperplane theorems
- Closest point theorems for convex sets

Today (Lecture 4): Convex Functions and Characterizations

- Definition of convex and concave functions
- Strictly convex and strictly concave functions
- First-order and second-order conditions for convexity
- Operations that preserve convexity
- Continuity of convex functions
- Epigraphs and hypographs

Next Time (Lecture 5): Subgradients

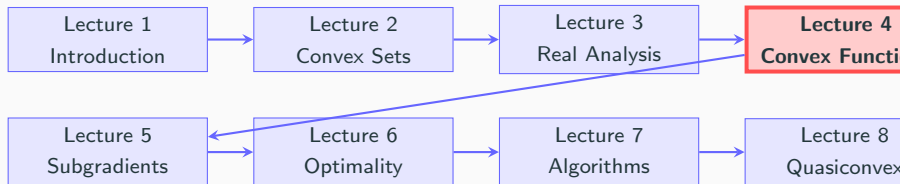
- Subgradients for non-differentiable convex functions
- Subdifferential: the set of all subgradients
- Existence of subgradients at interior points

Coming Soon

- **Lecture 6:** Optimality conditions for convex optimization
- **Lecture 7:** Algorithms (gradient descent, subgradient methods)
- **Lecture 8:** Quasiconvex and quasiconcave functions

Big Picture: We shift focus from convex *sets* to convex *functions*—the key to understanding optimization problems.

Big Picture: Course Roadmap



Why Convex Functions Matter

- Bridge from geometry (convex sets) to analysis (optimization)
- Convex functions + convex sets = tractable optimization
- Every local minimum is a global minimum for convex functions
- Foundation for duality theory and algorithms

Convex and Concave Functions

Strictly Convex and Strictly Concave Functions

Examples of (Strictly) Convex and Concave Functions

Characterizations of Convex Functions

Proving Convexity via Composition Rules

Calculus Background

First-Order Condition for Convexity

Second-Order Calculus Background

Second-Order Condition for Convexity

Operations Preserving Convexity of Functions

Continuity of Convex Functions

Epigraph and Hypograph

Roadmap

Convex and Concave Functions

Strictly Convex and Strictly Concave Functions

Examples of (Strictly) Convex and Concave Functions

Characterizations of Convex Functions

Proving Convexity via Composition Rules

Calculus Background

First-Order Condition for Convexity

Second-Order Calculus Background

Second-Order Condition for Convexity

Operations Preserving Convexity of Functions

Continuity of Convex Functions

Epigraph and Hypograph

Convex and Concave Functions

Let $X \subseteq \mathbb{R}^n$ be a convex set.

Convex Function: A function $f : X \rightarrow \mathbb{R}$ is said to be convex if for *any* two points $\mathbf{x}_1$ and $\mathbf{x}_2$ in X and *any* $\lambda \in [0, 1]$,

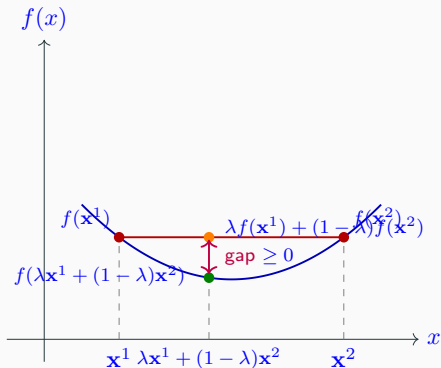
$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) \leq \lambda f(\mathbf{x}_1) + (1 - \lambda) f(\mathbf{x}_2).$$

Concave Function: A function $f : X \rightarrow \mathbb{R}$ is said to be concave if $-f$ is convex, i.e., for *any* two points $\mathbf{x}_1$ and $\mathbf{x}_2$ in X and *any* $\lambda \in [0, 1]$,

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) \geq \lambda f(\mathbf{x}_1) + (1 - \lambda) f(\mathbf{x}_2).$$

Only affine functions ($\mathbf{a}^T \mathbf{x} + b$) are *both* convex and concave.

Geometric Interpretation of Convexity



Key Insight: For a convex function, the graph lies *below* any chord connecting two points. The function value at a convex combination is at most the convex combination of function values.

Exercise: Verifying Convexity from the Definition

Quick Check

Using the definition, determine whether each function is convex, concave, both, or neither on the given domain:

1. $f(x) = 3x + 5$ on $X = \mathbb{R}$
2. $f(x) = x^2$ on $X = \mathbb{R}$
3. $f(x) = |x|$ on $X = \mathbb{R}$
4. $f(x) = \sin(x)$ on $X = [0, \pi]$

Hint: For each function, check if

$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$ for all x_1, x_2 and $\lambda \in [0, 1]$.

Solution: Verifying Convexity from the Definition

Answers

1. $f(x) = 3x + 5$: **Both convex and concave** (affine function).

$$f(\lambda x_1 + (1-\lambda)x_2) = 3(\lambda x_1 + (1-\lambda)x_2) + 5 = \lambda(3x_1 + 5) + (1-\lambda)(3x_2 + 5)$$

2. $f(x) = x^2$: **Convex**. We have:

$$(\lambda x_1 + (1-\lambda)x_2)^2 \leq \lambda x_1^2 + (1-\lambda)x_2^2$$

This follows since $\lambda(1-\lambda)(x_1 - x_2)^2 \geq 0$.

3. $f(x) = |x|$: **Convex**. By triangle inequality:

$$|\lambda x_1 + (1-\lambda)x_2| \leq \lambda|x_1| + (1-\lambda)|x_2|$$

4. $f(x) = \sin(x)$ on $[0, \pi]$: **Concave**. The sine function curves downward on this interval (second derivative $-\sin(x) \leq 0$).

Strictly Convex and Strictly Concave Functions

Let $X \subseteq \mathbb{R}^n$ be a convex set.

Strictly Convex Function: A function $f : X \rightarrow \mathbb{R}$ is said to be **strictly convex** if for *any* two **distinct** points $\mathbf{x}_1 \neq \mathbf{x}_2$ in X and *any* $\lambda \in (0, 1)$,

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) < \lambda f(\mathbf{x}_1) + (1 - \lambda) f(\mathbf{x}_2).$$

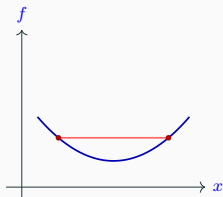
Strictly Concave Function: A function $f : X \rightarrow \mathbb{R}$ is said to be **strictly concave** if $-f$ is strictly convex, i.e., for *any* two **distinct** points $\mathbf{x}_1 \neq \mathbf{x}_2$ in X and *any* $\lambda \in (0, 1)$,

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) > \lambda f(\mathbf{x}_1) + (1 - \lambda) f(\mathbf{x}_2).$$

Key Differences from Regular Convexity

- Strict inequality ($<$ instead of $\leq$)
- Open interval $\lambda \in (0, 1)$ instead of closed $[0, 1]$
- Points must be distinct ($\mathbf{x}_1 \neq \mathbf{x}_2$)

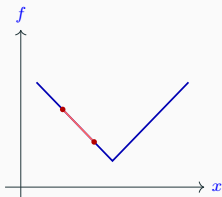
Strict vs Non-Strict Convexity



Strictly Convex

$$f(x) = x^2$$

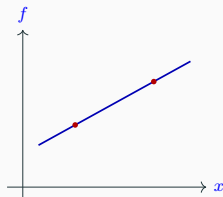
Strict gap always



Convex, Not Strictly

$$f(x) = |x|$$

Chord = graph on segments



Affine (Both)

$$f(x) = ax + b$$

Convex & Concave

Important Implications

- Strictly convex $\Rightarrow$ convex (but not vice versa)
- Affine functions are convex but **not** strictly convex
- Strict convexity guarantees **unique** global minimizers

Exercise: Strict Convexity

Think About It

Answer the following questions:

1. Are affine functions ($\mathbf{a}^T \mathbf{x} + b$) strictly convex? Strictly concave? Explain.
2. If f is strictly convex and has a minimizer on a convex set X , how many minimizers can it have?
3. Give an example of a function that is convex but not strictly convex.
4. Is the function $f(x, y) = x^2 + y^2$ strictly convex on $\mathbb{R}^2$?

Hint: Think about what “strict inequality” means geometrically.

Solution: Strict Convexity

Answers

1. **Affine functions:** Neither strictly convex nor strictly concave.

For any $\mathbf{x}_1 \neq \mathbf{x}_2$ and $\lambda \in (0, 1)$:

$$\mathbf{a}^T(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) + b = \lambda(\mathbf{a}^T \mathbf{x}_1 + b) + (1 - \lambda)(\mathbf{a}^T \mathbf{x}_2 + b)$$

This is equality, not strict inequality.

2. **Number of minimizers:** Exactly one (if it exists). If $\mathbf{x}_1$ and $\mathbf{x}_2$ were both minimizers with $\mathbf{x}_1 \neq \mathbf{x}_2$, then for $\lambda = 0.5$:

$$f(0.5\mathbf{x}_1 + 0.5\mathbf{x}_2) < 0.5f(\mathbf{x}_1) + 0.5f(\mathbf{x}_2) = f(\mathbf{x}_1)$$

contradicting that $\mathbf{x}_1$ is a minimizer.

3. **Convex but not strictly:** $f(x) = |x|$, $f(x) = \max(0, x)$, any piecewise linear convex function.
4. $f(x, y) = x^2 + y^2$: **Yes, strictly convex.** It's a sum of strictly convex functions.

Some Examples of Convex and Concave Functions

Examples of convex functions:

- Exponential: $\exp(x)$ on $X = \mathbb{R}$. (Strictly convex)
- Power functions: x^p on $X = (0, +\infty)$ for $p \geq 1$ or $p \leq 0$.
- Negative entropy: $x \ln(x)$ on $X = (0, +\infty)$. (Strictly convex)
- Norms: $\|\mathbf{x}\|_p$ on $X = \mathbb{R}^n$ for $p \geq 1$.
- Quadratic: $\mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ on $X = \mathbb{R}^n$ for $\mathbf{A} \succeq 0$. (Strictly convex if $\mathbf{A} \succ 0$)

Examples of concave functions:

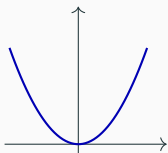
- Power functions: x^p on $X = (0, +\infty)$ for $p \in [0, 1]$.
- Geometric mean: $(\prod_{i=1}^n x_i)^{1/n}$ on $X = \mathbb{R}_{++}^n$.
- Logarithm: $\ln(x)$ on $X = (0, +\infty)$. (Strictly concave)
- Negative quadratic: $-\mathbf{x}^T \mathbf{A} \mathbf{x}$ for $\mathbf{A} \succeq 0$.

Note: $\mathbf{A} \succeq 0$ means $\mathbf{A}$ is positive semidefinite; $\mathbf{A} \succ 0$ means positive definite.

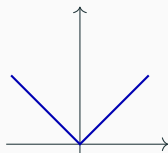
Visual Gallery: Convex and Concave Functions



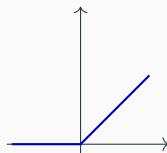
e^x (convex)



x^2 (str. convex)



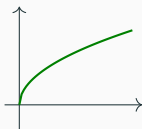
$|x|$ (convex)



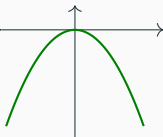
$\max(0, x)$ (convex)



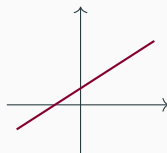
$\ln(x)$ (concave)



$\sqrt{x}$ (concave)



$-x^2$ (str. concave)



$ax + b$ (both)

Exercise: Classifying Functions

Classification Challenge

Classify each function as: convex, strictly convex, concave, strictly concave, both, or neither. Specify the domain.

Function	Convex?	Strictly?	Domain
$f(x) = 3x + 7$			
$f(x) = x^3$			
$f(x) = \max(x, 0)$			
$f(\mathbf{x}) = \ \mathbf{x}\ _2$			
$f(\mathbf{x}) = x_1^2 + x_2^2$			

Hint: Use the definition, known examples, or preservation rules.

Solution: Classifying Functions

Answers

Function	Convex?	Strictly?	Domain
$f(x) = 3x + 7$	Both	No	$\mathbb{R}$
$f(x) = x^3$	Neither	–	$\mathbb{R}$
$f(x) = \max(x, 0)$	Yes	No	$\mathbb{R}$
$f(\mathbf{x}) = \ \mathbf{x}\ _2$	Yes	No	$\mathbb{R}^n$
$f(\mathbf{x}) = x_1^2 + x_2^2$	Yes	Yes	$\mathbb{R}^n$

Explanations:

- $3x + 7$: Affine functions are both convex and concave, but not strictly either.
- x^3 : Check $x_1 = -1, x_2 = 1, \lambda = 0.5$: midpoint gives $f(0) = 0$, but average of endpoints is 0. Equality holds here, but fails convexity for other choices.
- $\max(x, 0)$: Max of two affine (convex) functions. Not strict (linear on $x \geq 0$).
- $\|\mathbf{x}\|_2$: Triangle inequality $\Rightarrow$ convex. Not strictly (linear along rays).
- $x_1^2 + x_2^2$: Sum of strictly convex functions (we proved x^2 is strictly convex).

Roadmap

Convex and Concave Functions

Strictly Convex and Strictly Concave Functions

Examples of (Strictly) Convex and Concave Functions

Characterizations of Convex Functions

Proving Convexity via Composition Rules

Calculus Background

First-Order Condition for Convexity

Second-Order Calculus Background

Second-Order Condition for Convexity

Operations Preserving Convexity of Functions

Continuity of Convex Functions

Epigraph and Hypograph

Proving Convexity: Multiple Approaches

Throughout this section, we'll use $f(x, y) = x^2 + 2y^2$ as a **running example** to demonstrate different ways to prove convexity:

1. **Composition Rules:** Build from known convex functions (we proved x^2 is convex)
2. **First-Order Condition:** For differentiable f , verify
$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x})$$
3. **Second-Order Condition:** For twice-differentiable f , verify
$$\nabla^2 f(\mathbf{x}) \succeq 0$$

Why Multiple Approaches?

- Composition rules: Quick when applicable, builds on known results
- First-order: Connects convexity to tangent planes, useful for optimality
- Second-order: Often the easiest verification for smooth functions

Running Example: Convexity via Composition Rules

Example: Prove $f(x, y) = x^2 + 2y^2$ is convex on $\mathbb{R}^2$

Approach: Use the fact that nonnegative weighted sums of convex functions are convex.

Running Example: Convexity via Composition Rules

Example: Prove $f(x, y) = x^2 + 2y^2$ is convex on $\mathbb{R}^2$

Approach: Use the fact that nonnegative weighted sums of convex functions are convex.

Step 1: We already proved $g(t) = t^2$ is convex on $\mathbb{R}$ (from definition).

Running Example: Convexity via Composition Rules

Example: Prove $f(x, y) = x^2 + 2y^2$ is convex on $\mathbb{R}^2$

Approach: Use the fact that nonnegative weighted sums of convex functions are convex.

Step 1: We already proved $g(t) = t^2$ is convex on $\mathbb{R}$ (from definition).

Step 2: Define $g_1(x, y) = x^2$ and $g_2(x, y) = y^2$.

These are compositions of t^2 with linear functions, so both are convex.

Running Example: Convexity via Composition Rules

Example: Prove $f(x, y) = x^2 + 2y^2$ is convex on $\mathbb{R}^2$

Approach: Use the fact that nonnegative weighted sums of convex functions are convex.

Step 1: We already proved $g(t) = t^2$ is convex on $\mathbb{R}$ (from definition).

Step 2: Define $g_1(x, y) = x^2$ and $g_2(x, y) = y^2$.

These are compositions of t^2 with linear functions, so both are convex.

Step 3: Write $f(x, y) = 1 \cdot g_1(x, y) + 2 \cdot g_2(x, y)$.

This is a nonnegative weighted sum (weights $1 \geq 0$ and $2 \geq 0$).

Running Example: Convexity via Composition Rules

Example: Prove $f(x, y) = x^2 + 2y^2$ is convex on $\mathbb{R}^2$

Approach: Use the fact that nonnegative weighted sums of convex functions are convex.

Step 1: We already proved $g(t) = t^2$ is convex on $\mathbb{R}$ (from definition).

Step 2: Define $g_1(x, y) = x^2$ and $g_2(x, y) = y^2$.

These are compositions of t^2 with linear functions, so both are convex.

Step 3: Write $f(x, y) = 1 \cdot g_1(x, y) + 2 \cdot g_2(x, y)$.

This is a nonnegative weighted sum (weights $1 \geq 0$ and $2 \geq 0$).

Conclusion

By the preservation rule for nonnegative weighted sums:

$$f(x, y) = x^2 + 2y^2 \text{ is convex. } \square$$

Note: Since x^2 and y^2 are strictly convex, the sum is also **strictly convex**.

Calculus Review: Derivatives

For characterizing convexity of differentiable functions, we need:

Derivative (Scalar Function)

For $f : \mathbb{R} \rightarrow \mathbb{R}$, the derivative at x is:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Example: $f(x) = x^2 \implies f'(x) = 2x$

Calculus Review: Derivatives

For characterizing convexity of differentiable functions, we need:

Derivative (Scalar Function)

For $f : \mathbb{R} \rightarrow \mathbb{R}$, the derivative at x is:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Example: $f(x) = x^2 \implies f'(x) = 2x$

Partial Derivative (Multivariate Function)

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the partial derivative with respect to x_i is:

$$\frac{\partial f}{\partial x_i}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h}$$

where $\mathbf{e}_i$ is the i -th standard basis vector.

Example: $f(x, y) = x^2 + 2y^2 \implies \frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 4y$

Calculus Review: Gradient

Gradient

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable, the **gradient** is the vector of all partial derivatives:

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\mathbf{x}) \end{pmatrix} \in \mathbb{R}^n$$

Calculus Review: Gradient

Gradient

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable, the **gradient** is the vector of all partial derivatives:

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\mathbf{x}) \end{pmatrix} \in \mathbb{R}^n$$

Example: For $f(x, y) = x^2 + 2y^2$:

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

At the point $(1, 2)$: $\nabla f(1, 2) = \begin{pmatrix} 2 \\ 8 \end{pmatrix}$

Calculus Review: Gradient

Gradient

For $f: \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable, the **gradient** is the vector of all partial derivatives:

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\mathbf{x}) \end{pmatrix} \in \mathbb{R}^n$$

Example: For $f(x, y) = x^2 + 2y^2$:

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

At the point $(1, 2)$: $\nabla f(1, 2) = \begin{pmatrix} 2 \\ 8 \end{pmatrix}$

Geometric Meaning

- $\nabla f(\mathbf{x})$ points in the direction of **steepest increase** of f at $\mathbf{x}$
- $\|\nabla f(\mathbf{x})\|$ gives the rate of increase in that direction
- $\nabla f(\mathbf{x}) = \mathbf{0}$ at critical points (candidates for min/max)

Calculus Review: Directional Derivative

Directional Derivative

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable at $\mathbf{x}$, the directional derivative in direction

$\mathbf{d} \in \mathbb{R}^n$ is:

$$D_{\mathbf{d}}f(\mathbf{x}) = \lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{d}) - f(\mathbf{x})}{t} = \nabla f(\mathbf{x})^T \mathbf{d}$$

Calculus Review: Directional Derivative

Directional Derivative

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable at $\mathbf{x}$, the directional derivative in direction $\mathbf{d} \in \mathbb{R}^n$ is:

$$D_{\mathbf{d}}f(\mathbf{x}) = \lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{d}) - f(\mathbf{x})}{t} = \nabla f(\mathbf{x})^T \mathbf{d}$$

Example: For $f(x, y) = x^2 + 2y^2$ at point $(1, 1)$ in direction $\mathbf{d} = (1, 1)^T$:

$$\nabla f(1, 1) = \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \quad D_{\mathbf{d}}f(1, 1) = \begin{pmatrix} 2 \\ 4 \end{pmatrix}^T \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2 + 4 = 6$$

Calculus Review: Directional Derivative

Directional Derivative

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable at $\mathbf{x}$, the **directional derivative** in direction $\mathbf{d} \in \mathbb{R}^n$ is:

$$D_{\mathbf{d}}f(\mathbf{x}) = \lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{d}) - f(\mathbf{x})}{t} = \nabla f(\mathbf{x})^T \mathbf{d}$$

Example: For $f(x, y) = x^2 + 2y^2$ at point $(1, 1)$ in direction $\mathbf{d} = (1, 1)^T$:

$$\nabla f(1, 1) = \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \quad D_{\mathbf{d}}f(1, 1) = \begin{pmatrix} 2 \\ 4 \end{pmatrix}^T \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2 + 4 = 6$$

Connection to First-Order Condition

The first-order Taylor approximation of f near $\mathbf{x}$ is:

$$f(\mathbf{x} + \mathbf{d}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{d}$$

For convex f , this linear approximation is a **global underestimator**.

First-Order Condition for Convexity

Theorem (First-Order Condition)

Let $S \subseteq \mathbb{R}^n$ be a nonempty open convex set, and let $f : S \rightarrow \mathbb{R}$ be differentiable on S . Then f is convex on S if and only if

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) \quad \text{for all } \mathbf{x}, \mathbf{y} \in S.$$

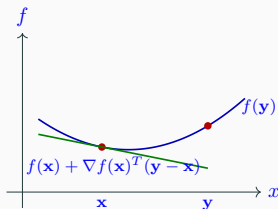
First-Order Condition for Convexity

Theorem (First-Order Condition)

Let $S \subseteq \mathbb{R}^n$ be a nonempty open convex set, and let $f : S \rightarrow \mathbb{R}$ be differentiable on S . Then f is convex on S if and only if

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) \quad \text{for all } \mathbf{x}, \mathbf{y} \in S.$$

Geometric Interpretation:



The graph of a convex function lies **above** all its tangent lines/planes.

Proof of First-Order Condition

Theorem

f differentiable on open convex S is convex $\Leftrightarrow$

$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in S$.

Proof of First-Order Condition

Theorem

f differentiable on open convex S is convex $\Leftrightarrow$

$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in S$.

Proof ($\Rightarrow$): Assume f is convex. For $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in (0, 1]$:

$$f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) \leq (1 - \lambda)f(\mathbf{x}) + \lambda f(\mathbf{y})$$

Rearranging:
$$\frac{f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\lambda} \leq f(\mathbf{y}) - f(\mathbf{x})$$

Proof of First-Order Condition

Theorem

f differentiable on open convex S is convex $\Leftrightarrow$

$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in S$.

Proof ($\Rightarrow$): Assume f is convex. For $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in (0, 1]$:

$$f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) \leq (1 - \lambda)f(\mathbf{x}) + \lambda f(\mathbf{y})$$

Rearranging: $\frac{f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\lambda} \leq f(\mathbf{y}) - f(\mathbf{x})$

Taking $\lambda \rightarrow 0^+$: $\nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) - f(\mathbf{x})$

✓

Proof of First-Order Condition

Theorem

f differentiable on open convex S is convex $\Leftrightarrow$

$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in S$.

Proof ($\Rightarrow$): Assume f is convex. For $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in (0, 1]$:

$$f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) \leq (1 - \lambda)f(\mathbf{x}) + \lambda f(\mathbf{y})$$

Rearranging: $\frac{f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\lambda} \leq f(\mathbf{y}) - f(\mathbf{x})$

Taking $\lambda \rightarrow 0^+$: $\nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) - f(\mathbf{x})$ ✓

Proof ($\Leftarrow$): Assume the first-order condition holds. For $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$, let $\mathbf{z} = \lambda\mathbf{x} + (1 - \lambda)\mathbf{y}$. Then:

$$f(\mathbf{x}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^T(\mathbf{x} - \mathbf{z})$$

$$f(\mathbf{y}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^T(\mathbf{y} - \mathbf{z})$$

Proof of First-Order Condition

Theorem

f differentiable on open convex S is convex $\Leftrightarrow$

$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in S$.

Proof ($\Rightarrow$): Assume f is convex. For $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in (0, 1]$:

$$f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) \leq (1 - \lambda)f(\mathbf{x}) + \lambda f(\mathbf{y})$$

Rearranging: $\frac{f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\lambda} \leq f(\mathbf{y}) - f(\mathbf{x})$

Taking $\lambda \rightarrow 0^+$: $\nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) - f(\mathbf{x})$ ✓

Proof ($\Leftarrow$): Assume the first-order condition holds. For $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$, let $\mathbf{z} = \lambda\mathbf{x} + (1 - \lambda)\mathbf{y}$. Then:

$$f(\mathbf{x}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^T(\mathbf{x} - \mathbf{z})$$

$$f(\mathbf{y}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^T(\mathbf{y} - \mathbf{z})$$

Multiply first by λ , second by $(1 - \lambda)$, add: $\lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}) \geq f(\mathbf{z})$

□

Running Example: First-Order Proof

Example: Prove $f(x, y) = x^2 + 2y^2$ is convex using the first-order condition

Step 1: Compute the gradient.

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

Running Example: First-Order Proof

Example: Prove $f(x, y) = x^2 + 2y^2$ is convex using the first-order condition

Step 1: Compute the gradient.

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

Step 2: The first-order condition requires: for all $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$,

$$f(x_2, y_2) \geq f(x_1, y_1) + \nabla f(x_1, y_1)^T \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$$

Running Example: First-Order Proof

Example: Prove $f(x, y) = x^2 + 2y^2$ is convex using the first-order condition

Step 1: Compute the gradient.

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

Step 2: The first-order condition requires: for all $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$,

$$f(x_2, y_2) \geq f(x_1, y_1) + \nabla f(x_1, y_1)^T \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$$

Step 3: Expand both sides.

LHS: $x_2^2 + 2y_2^2$

RHS:

$$x_1^2 + 2y_1^2 + 2x_1(x_2 - x_1) + 4y_1(y_2 - y_1) = -x_1^2 + 2x_1x_2 - 2y_1^2 + 4y_1y_2$$

Running Example: First-Order Proof

Example: Prove $f(x, y) = x^2 + 2y^2$ is convex using the first-order condition

Step 1: Compute the gradient.

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

Step 2: The first-order condition requires: for all $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$,

$$f(x_2, y_2) \geq f(x_1, y_1) + \nabla f(x_1, y_1)^T \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$$

Step 3: Expand both sides.

LHS: $x_2^2 + 2y_2^2$

RHS:

$$x_1^2 + 2y_1^2 + 2x_1(x_2 - x_1) + 4y_1(y_2 - y_1) = -x_1^2 + 2x_1x_2 - 2y_1^2 + 4y_1y_2$$

Step 4: Show $\text{LHS} - \text{RHS} \geq 0$.

$$x_2^2 + 2y_2^2 + x_1^2 - 2x_1x_2 + 2y_1^2 - 4y_1y_2 = (x_2 - x_1)^2 + 2(y_2 - y_1)^2 \geq 0 \quad \checkmark$$

□

Calculus Review: Hessian Matrix

Hessian

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ twice differentiable, the **Hessian** is the matrix of second partial derivatives:

$$\nabla^2 f(\mathbf{x}) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix} \in \mathbb{R}^{n \times n}$$

Calculus Review: Hessian Matrix

Hessian

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ twice differentiable, the **Hessian** is the matrix of second partial derivatives:

$$\nabla^2 f(\mathbf{x}) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix} \in \mathbb{R}^{n \times n}$$

Example: For $f(x, y) = x^2 + 2y^2$:

$$\frac{\partial^2 f}{\partial x^2} = 2, \quad \frac{\partial^2 f}{\partial y^2} = 4, \quad \frac{\partial^2 f}{\partial x \partial y} = 0$$

$$\nabla^2 f(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

Calculus Review: Hessian Matrix

Hessian

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ twice differentiable, the **Hessian** is the matrix of second partial derivatives:

$$\nabla^2 f(\mathbf{x}) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix} \in \mathbb{R}^{n \times n}$$

Example: For $f(x, y) = x^2 + 2y^2$:

$$\frac{\partial^2 f}{\partial x^2} = 2, \quad \frac{\partial^2 f}{\partial y^2} = 4, \quad \frac{\partial^2 f}{\partial x \partial y} = 0$$

$$\nabla^2 f(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

Note: For smooth functions, $\nabla^2 f$ is **symmetric** (mixed partials are equal).

Positive Semidefinite and Positive Definite Matrices

Definitions

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is:

- Positive semidefinite ($A \succeq 0$) if $\mathbf{d}^T A \mathbf{d} \geq 0$ for all $\mathbf{d} \in \mathbb{R}^n$
- Positive definite ($A \succ 0$) if $\mathbf{d}^T A \mathbf{d} > 0$ for all $\mathbf{d} \neq \mathbf{0}$

Positive Semidefinite and Positive Definite Matrices

Definitions

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is:

- **Positive semidefinite** ($A \succeq 0$) if $\mathbf{d}^T A \mathbf{d} \geq 0$ for all $\mathbf{d} \in \mathbb{R}^n$
- **Positive definite** ($A \succ 0$) if $\mathbf{d}^T A \mathbf{d} > 0$ for all $\mathbf{d} \neq \mathbf{0}$

Equivalent Conditions (any one implies the others):

1. All eigenvalues of A are ≥ 0 (resp. > 0)
2. All leading principal minors are ≥ 0 (resp. > 0 for PD)
3. $A = B^T B$ for some matrix B (resp. B has full column rank)

Positive Semidefinite and Positive Definite Matrices

Definitions

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is:

- **Positive semidefinite** ($A \succeq 0$) if $\mathbf{d}^T A \mathbf{d} \geq 0$ for all $\mathbf{d} \in \mathbb{R}^n$
- **Positive definite** ($A \succ 0$) if $\mathbf{d}^T A \mathbf{d} > 0$ for all $\mathbf{d} \neq 0$

Equivalent Conditions (any one implies the others):

1. All eigenvalues of A are ≥ 0 (resp. > 0)
2. All leading principal minors are ≥ 0 (resp. > 0 for PD)
3. $A = B^T B$ for some matrix B (resp. B has full column rank)

Example: For $\nabla^2 f = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$:

Eigenvalues are 2 and 4, both $> 0 \Rightarrow \nabla^2 f \succ 0$ (positive definite)

Second-Order Condition for Convexity

Theorem (Second-Order Condition)

Let $S \subseteq \mathbb{R}^n$ be a nonempty open convex set, and let $f : S \rightarrow \mathbb{R}$ be twice continuously differentiable on S . Then:

1. f is convex on $S \Leftrightarrow \nabla^2 f(\mathbf{x}) \succeq 0$ for all $\mathbf{x} \in S$ (positive semidefinite)
2. f is **strictly** convex on $S \Leftrightarrow \nabla^2 f(\mathbf{x}) \succ 0$ for all $\mathbf{x} \in S$ (positive definite)

Second-Order Condition for Convexity

Theorem (Second-Order Condition)

Let $S \subseteq \mathbb{R}^n$ be a nonempty open convex set, and let $f : S \rightarrow \mathbb{R}$ be twice continuously differentiable on S . Then:

1. f is convex on $S \Leftrightarrow \nabla^2 f(\mathbf{x}) \succeq 0$ for all $\mathbf{x} \in S$ (positive semidefinite)
2. f is **strictly** convex on $S \Leftarrow \nabla^2 f(\mathbf{x}) \succ 0$ for all $\mathbf{x} \in S$ (positive definite)

Recall: For $f : \mathbb{R} \rightarrow \mathbb{R}$, the Hessian is just $f''(x)$. So:

- f convex $\Leftrightarrow f''(x) \geq 0$ for all x
- f strictly convex $\Leftarrow f''(x) > 0$ for all x

Second-Order Condition for Convexity

Theorem (Second-Order Condition)

Let $S \subseteq \mathbb{R}^n$ be a nonempty open convex set, and let $f : S \rightarrow \mathbb{R}$ be **twice continuously differentiable** on S . Then:

1. f is convex on $S \Leftrightarrow \nabla^2 f(\mathbf{x}) \succeq 0$ for all $\mathbf{x} \in S$ (positive semidefinite)
2. f is **strictly** convex on $S \Leftarrow \nabla^2 f(\mathbf{x}) \succ 0$ for all $\mathbf{x} \in S$ (positive definite)

Recall: For $f : \mathbb{R} \rightarrow \mathbb{R}$, the Hessian is just $f''(x)$. So:

- f convex $\Leftrightarrow f''(x) \geq 0$ for all x
- f strictly convex $\Leftarrow f''(x) > 0$ for all x

Important Note on Strict Convexity

The converse of (2) is **not** true! Example: $f(x) = x^4$ is strictly convex, but $f''(0) = 0$.

Proof Sketch: Second-Order Condition

Theorem

f twice differentiable on open convex S is convex $\Leftrightarrow \nabla^2 f(\mathbf{x}) \succeq 0$ for all $\mathbf{x} \in S$.

Proof Sketch: Second-Order Condition

Theorem

f twice differentiable on open convex S is convex $\Leftrightarrow \nabla^2 f(\mathbf{x}) \succeq 0$ for all $\mathbf{x} \in S$.

Proof ($\Rightarrow$): Assume f is convex. By first-order condition:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) \quad \text{for all } \mathbf{x}, \mathbf{y}$$

Let $\mathbf{y} = \mathbf{x} + t\mathbf{d}$ for small $t > 0$ and any direction $\mathbf{d}$. Taylor expansion:

$$f(\mathbf{x} + t\mathbf{d}) = f(\mathbf{x}) + t\nabla f(\mathbf{x})^T \mathbf{d} + \frac{t^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}) \mathbf{d} + o(t^2)$$

Proof Sketch: Second-Order Condition

Theorem

f twice differentiable on open convex S is convex $\Leftrightarrow \nabla^2 f(\mathbf{x}) \succeq 0$ for all $\mathbf{x} \in S$.

Proof ($\Rightarrow$): Assume f is convex. By first-order condition:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) \quad \text{for all } \mathbf{x}, \mathbf{y}$$

Let $\mathbf{y} = \mathbf{x} + t\mathbf{d}$ for small $t > 0$ and any direction $\mathbf{d}$. Taylor expansion:

$$f(\mathbf{x} + t\mathbf{d}) = f(\mathbf{x}) + t\nabla f(\mathbf{x})^T \mathbf{d} + \frac{t^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}) \mathbf{d} + o(t^2)$$

Substituting into first-order condition and dividing by $\frac{t^2}{2}$:

$$\mathbf{d}^T \nabla^2 f(\mathbf{x}) \mathbf{d} \geq 0 \quad \text{for all } \mathbf{d}$$

This means $\nabla^2 f(\mathbf{x}) \succeq 0$.

✓

Proof Sketch: Second-Order Condition

Theorem

f twice differentiable on open convex S is convex $\Leftrightarrow \nabla^2 f(\mathbf{x}) \succeq 0$ for all $\mathbf{x} \in S$.

Proof ($\Rightarrow$): Assume f is convex. By first-order condition:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) \quad \text{for all } \mathbf{x}, \mathbf{y}$$

Let $\mathbf{y} = \mathbf{x} + t\mathbf{d}$ for small $t > 0$ and any direction $\mathbf{d}$. Taylor expansion:

$$f(\mathbf{x} + t\mathbf{d}) = f(\mathbf{x}) + t\nabla f(\mathbf{x})^T \mathbf{d} + \frac{t^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}) \mathbf{d} + o(t^2)$$

Substituting into first-order condition and dividing by $\frac{t^2}{2}$:

$$\mathbf{d}^T \nabla^2 f(\mathbf{x}) \mathbf{d} \geq 0 \quad \text{for all } \mathbf{d}$$

This means $\nabla^2 f(\mathbf{x}) \succeq 0$. ✓

Proof ($\Leftarrow$): Uses Taylor expansion with integral form of remainder. See Bazaraa et al. Theorem 3.3.8. □

Running Example: Second-Order Proof

Example: Prove $f(x, y) = x^2 + 2y^2$ is convex using the second-order condition

Step 1: Compute the gradient.

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

Running Example: Second-Order Proof

Example: Prove $f(x, y) = x^2 + 2y^2$ is convex using the second-order condition

Step 1: Compute the gradient.

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

Step 2: Compute the Hessian.

$$\nabla^2 f(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

Running Example: Second-Order Proof

Example: Prove $f(x, y) = x^2 + 2y^2$ is convex using the second-order condition

Step 1: Compute the gradient.

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

Step 2: Compute the Hessian.

$$\nabla^2 f(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

Step 3: Check positive definiteness.

Eigenvalues: $\lambda_1 = 2 > 0$, $\lambda_2 = 4 > 0$

Running Example: Second-Order Proof

Example: Prove $f(x, y) = x^2 + 2y^2$ is convex using the second-order condition

Step 1: Compute the gradient.

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

Step 2: Compute the Hessian.

$$\nabla^2 f(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

Step 3: Check positive definiteness.

Eigenvalues: $\lambda_1 = 2 > 0$, $\lambda_2 = 4 > 0$

Since $\nabla^2 f \succ 0$, $f(x, y) = x^2 + 2y^2$ is **strictly** convex. □

Comparison of Methods for $f(x, y) = x^2 + 2y^2$

Method	Key Step	Result
Composition	Sum of convex functions	Strictly convex
First-Order	Show $(x_2 - x_1)^2 + 2(y_2 - y_1)^2 \geq 0$	Convex
Second-Order	Eigenvalues of $\nabla^2 f$ are $2, 4 > 0$	Strictly convex

Second-Order Condition: Multivariate Example

Example: Is $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ convex?

Second-Order Condition: Multivariate Example

Example: Is $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ convex?

Step 1: Compute the gradient.

$$\nabla f(\mathbf{x}) = 2A\mathbf{x} + \mathbf{b} \quad (\text{assuming } A \text{ symmetric})$$

Second-Order Condition: Multivariate Example

Example: Is $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ convex?

Step 1: Compute the gradient.

$$\nabla f(\mathbf{x}) = 2A\mathbf{x} + \mathbf{b} \quad (\text{assuming } A \text{ symmetric})$$

Step 2: Compute the Hessian.

$$\nabla^2 f(\mathbf{x}) = 2A$$

Second-Order Condition: Multivariate Example

Example: Is $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ convex?

Step 1: Compute the gradient.

$$\nabla f(\mathbf{x}) = 2A\mathbf{x} + \mathbf{b} \quad (\text{assuming } A \text{ symmetric})$$

Step 2: Compute the Hessian.

$$\nabla^2 f(\mathbf{x}) = 2A$$

Step 3: Check positive semidefiniteness.

$$\nabla^2 f(\mathbf{x}) = 2A \succeq 0 \Leftrightarrow A \succeq 0 \quad (\text{i.e., } A \text{ is positive semidefinite})$$

Second-Order Condition: Multivariate Example

Example: Is $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ convex?

Step 1: Compute the gradient.

$$\nabla f(\mathbf{x}) = 2A\mathbf{x} + \mathbf{b} \quad (\text{assuming } A \text{ symmetric})$$

Step 2: Compute the Hessian.

$$\nabla^2 f(\mathbf{x}) = 2A$$

Step 3: Check positive semidefiniteness.

$$\nabla^2 f(\mathbf{x}) = 2A \succeq 0 \Leftrightarrow A \succeq 0 \quad (\text{i.e., } A \text{ is positive semidefinite})$$

Conclusion

- $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ is convex $\Leftrightarrow A \succeq 0$
- f is **strictly** convex $\Leftrightarrow A \succ 0$

This is one of the most important examples in optimization!

Exercise: Using First and Second-Order Conditions

Practice

For each function, use the second-order condition to determine convexity:

1. $f(x) = e^x$ on $\mathbb{R}$
2. $f(x) = -\ln(x)$ on $(0, \infty)$
3. $f(x, y) = x^2 + xy + y^2$ on $\mathbb{R}^2$
4. $f(x, y) = x^2 - y^2$ on $\mathbb{R}^2$

Hint: For multivariate functions, check if the Hessian is positive semidefinite.

Solution: Using First and Second-Order Conditions

Answers

1. $f(x) = e^x$: $f''(x) = e^x > 0$ for all x . **Strictly convex.**
2. $f(x) = -\ln(x)$: $f'(x) = -1/x$, $f''(x) = 1/x^2 > 0$ for $x > 0$.
Strictly convex.
3. $f(x, y) = x^2 + xy + y^2$:

$$\nabla^2 f = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Eigenvalues: $\lambda = 3, 1 > 0$. **Strictly convex.**

4. $f(x, y) = x^2 - y^2$:

$$\nabla^2 f = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$$

Eigenvalues: $\lambda = 2, -2$. Indefinite. **Neither convex nor concave.**

Operations Preserving Convexity of Functions

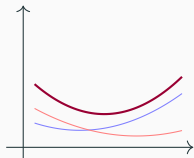
Suppose $f, f_1, \dots, f_m : \mathbb{R}^n \rightarrow \mathbb{R}$ are convex. The following are convex:

1. **Nonnegative Weighted Sum:** $g(\mathbf{x}) = \sum_{i=1}^m \alpha_i f_i(\mathbf{x})$ for $\alpha_i \geq 0$.
2. **Pointwise Maximum:** $g(\mathbf{x}) = \max_{i \in [m]} f_i(\mathbf{x})$.
3. **Affine Composition:** $g(\mathbf{x}) = f(A\mathbf{x} + \mathbf{b})$ for any $A \in \mathbb{R}^{k \times n}$, $\mathbf{b} \in \mathbb{R}^k$.
4. **Composition Rules:** $h(\mathbf{x}) = g(f(\mathbf{x}))$ where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$:
 - If f convex and g convex & nondecreasing $\Rightarrow h$ convex
 - If f concave and g convex & nonincreasing $\Rightarrow h$ convex
5. **Partial Minimization:** If $f(\mathbf{x}, \mathbf{y})$ is convex and C is convex, then

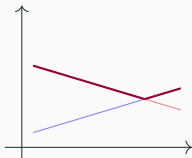
$$g(\mathbf{x}) = \inf_{\mathbf{y} \in C} f(\mathbf{x}, \mathbf{y})$$

is convex (assuming the infimum is achieved).

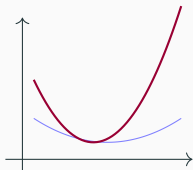
Visualizing Convexity-Preserving Operations



Sum: $f_1 + f_2$



Max: $\max(f_1, f_2)$



Affine: $f(Ax + b)$

Why These Rules Matter

- Build complex convex functions from simple ones
- Verify convexity without computing second derivatives
- Essential for recognizing convex optimization problems

Exercise: Using Convexity Preservation Rules

Apply the Rules

For each function, determine if it is convex by identifying which preservation rule(s) apply:

1. $f(\mathbf{x}) = \max\{x_1 + 2x_2, 3x_1 - x_2, -x_1 + x_2 + 5\}$

2. $f(x_1, x_2) = e^{2x_1 - x_2} + (x_1 + 3x_2)^2$

3. $f(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{b}\|_2^2$ where $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$

4. $f(x) = e^{x^2}$

Hint: Break each function into compositions of simpler functions.

Solution: Using Convexity Preservation Rules

Answers

1. $\max\{x_1 + 2x_2, 3x_1 - x_2, -x_1 + x_2 + 5\}$: **Convex**

Pointwise maximum of affine (hence convex) functions.

2. $e^{2x_1 - x_2} + (x_1 + 3x_2)^2$: **Convex**

$2x_1 - x_2$ is affine and e^t is convex $\Rightarrow e^{2x_1 - x_2}$ is convex (affine composition). Similarly, $x_1 + 3x_2$ is affine and t^2 is convex $\Rightarrow (x_1 + 3x_2)^2$ is convex. Sum of convex is convex.

3. $\|Ax - b\|_2^2$: **Convex**

$\|\cdot\|_2^2$ is convex; $Ax - b$ is affine; composition of convex with affine is convex.

4. e^{x^2} : **Convex**

x^2 is convex; e^t is convex and nondecreasing; composition of convex nondecreasing with convex is convex.

Caution: Log-Sum-Exp and Composition Rules

Consider the log-sum-exp function:

$$f(\mathbf{x}) = \log(e^{x_1} + e^{x_2} + e^{x_3})$$

Tempting (but wrong) argument:

- Each e^{x_i} is convex; sum of convex is convex ✓
- $\log$ is increasing ✓
- So $\log \circ (\text{convex})$ is convex? **NO!**

Caution: Log-Sum-Exp and Composition Rules

Consider the **log-sum-exp** function:

$$f(\mathbf{x}) = \log(e^{x_1} + e^{x_2} + e^{x_3})$$

Tempting (but wrong) argument:

- Each e^{x_i} is convex; sum of convex is convex ✓
- $\log$ is increasing ✓
- So $\log \circ (\text{convex})$ is convex? **NO!**

Why This Fails

The composition rule requires g convex *and* nondecreasing. But $\log$ is **concave**, not convex. The rule “concave nondecreasing $\circ$ convex” does **not** preserve convexity.

Caution: Log-Sum-Exp and Composition Rules

Consider the **log-sum-exp** function:

$$f(\mathbf{x}) = \log(e^{x_1} + e^{x_2} + e^{x_3})$$

Tempting (but wrong) argument:

- Each e^{x_i} is convex; sum of convex is convex ✓
- $\log$ is increasing ✓
- So $\log \circ (\text{convex})$ is convex? **NO!**

Why This Fails

The composition rule requires g convex *and* nondecreasing. But $\log$ is **concave**, not convex. The rule “concave nondecreasing $\circ$ convex” does **not** preserve convexity.

Counterexample: Take $g(u) = u^2$ (convex). Then $\log(g(u)) = \log(u^2) = 2 \log |u|$, which is **not convex** on $\mathbb{R}$.

So how do we prove log-sum-exp is convex? We need a different technique ...

Proving Log-Sum-Exp is Convex via the Epigraph

Key Idea: Show the epigraph of f is convex using composition rules that *do* work.

Step 1: Rewrite the epigraph.

$t \geq f(\mathbf{x})$ means:

$$t \geq \log \left(\sum_{i=1}^3 e^{x_i} \right) \iff e^t \geq \sum_{i=1}^3 e^{x_i} \iff \sum_{i=1}^3 e^{x_i - t} \leq 1$$

Proving Log-Sum-Exp is Convex via the Epigraph

Key Idea: Show the epigraph of f is convex using composition rules that *do* work.

Step 1: Rewrite the epigraph.

$t \geq f(\mathbf{x})$ means:

$$t \geq \log\left(\sum_{i=1}^3 e^{x_i}\right) \iff e^t \geq \sum_{i=1}^3 e^{x_i} \iff \sum_{i=1}^3 e^{x_i - t} \leq 1$$

So $\text{epi}(f) = \left\{(\mathbf{x}, t) : \sum_{i=1}^3 e^{x_i - t} \leq 1\right\}$.

Step 2: Show the defining function is convex.

Define $\phi(\mathbf{x}, t) = \sum_{i=1}^3 e^{x_i - t}$. Each $x_i - t$ is **affine** in $(\mathbf{x}, t)$, $u \mapsto e^u$ is **convex**, so $e^{x_i - t}$ is convex by **affine composition**. Sum of convex $\Rightarrow \phi$ is **convex**.

Proving Log-Sum-Exp is Convex via the Epigraph

Key Idea: Show the epigraph of f is convex using composition rules that *do* work.

Step 1: Rewrite the epigraph.

$t \geq f(\mathbf{x})$ means:

$$t \geq \log\left(\sum_{i=1}^3 e^{x_i}\right) \iff e^t \geq \sum_{i=1}^3 e^{x_i} \iff \sum_{i=1}^3 e^{x_i - t} \leq 1$$

So $\text{epi}(f) = \left\{(\mathbf{x}, t) : \sum_{i=1}^3 e^{x_i - t} \leq 1\right\}$.

Step 2: Show the defining function is convex.

Define $\phi(\mathbf{x}, t) = \sum_{i=1}^3 e^{x_i - t}$. Each $x_i - t$ is **affine** in $(\mathbf{x}, t)$, $u \mapsto e^u$ is **convex**, so $e^{x_i - t}$ is convex by **affine composition**. Sum of convex $\Rightarrow \phi$ is **convex**.

Step 3: Sublevel sets of convex functions are convex.

$\text{epi}(f) = \{(\mathbf{x}, t) : \phi(\mathbf{x}, t) \leq 1\}$ is a sublevel set of the convex function ϕ , hence **convex**.

Since $\text{epi}(f)$ is convex, f is convex. □

The same argument proves $f(\mathbf{x}) = \log\left(\sum_{i=1}^n e^{x_i}\right)$ is convex for any n .

Application: Convexity of Value Functions

Question

Suppose $\mathbf{g} \in \mathbb{R}^p$, $A \in \mathbb{R}^{m \times p}$, $\mathbf{b} \in \mathbb{R}^m$, $G \in \mathbb{R}^{m \times n}$. Define the value function:

$$f(\mathbf{x}) := \min\{\mathbf{g}^T \mathbf{y} : A\mathbf{y} \geq \mathbf{b} - G\mathbf{x}, \mathbf{y} \in \mathbb{R}^p\} \text{ for each } \mathbf{x} \in \mathbb{R}^n.$$

Assume that for each $\mathbf{x}$, an optimal solution exists. Is f convex on $\mathbb{R}^n$?

Answer: Yes! This follows from the partial minimization rule.

- The objective $\mathbf{g}^T \mathbf{y}$ is affine (hence convex) in $(\mathbf{x}, \mathbf{y})$
- The constraint $A\mathbf{y} \geq \mathbf{b} - G\mathbf{x}$ defines a convex set in $(\mathbf{x}, \mathbf{y})$
- Minimizing a convex function over a convex set preserves convexity

This is fundamental in stochastic programming and bilevel optimization!

Roadmap

Convex and Concave Functions

Strictly Convex and Strictly Concave Functions

Examples of (Strictly) Convex and Concave Functions

Characterizations of Convex Functions

Proving Convexity via Composition Rules

Calculus Background

First-Order Condition for Convexity

Second-Order Calculus Background

Second-Order Condition for Convexity

Operations Preserving Convexity of Functions

Continuity of Convex Functions

Epigraph and Hypograph

Continuity of Convex Functions

Theorem (Continuity of Convex Functions)

Let $S \subseteq \mathbb{R}^n$ be a nonempty convex set, and $f : S \rightarrow \mathbb{R}$ be a convex function. Then f is continuous on the interior of S .

Proof: See Theorem 3.1.3 of Bazaraa et al. (reading assignment)

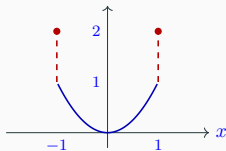
Key insight: Convexity implies a strong form of regularity. Convex functions cannot have “wild” oscillations in the interior.

Boundary behavior: Points of discontinuity are only allowed at the boundary of S .

Examples: Convex Functions with Boundary Discontinuities

Example 1: $S := \{x \in \mathbb{R} : -1 \leq x \leq 1\}$,

$$f(x) = \begin{cases} x^2 & \text{for } |x| < 1 \\ 2 & \text{for } |x| = 1 \end{cases}$$



Example 2: $S := \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_2 \leq 1\}$,

$$f(\mathbf{x}) = \begin{cases} 0 & \text{for } \|\mathbf{x}\|_2 < 1 \\ 1 & \text{for } \|\mathbf{x}\|_2 = 1 \end{cases}$$

Both are convex but discontinuous at boundary points.

Exercise: Continuity of Convex Functions

True or False

Determine whether each statement is true or false. Justify your answer.

1. Every convex function on $\mathbb{R}^n$ is continuous everywhere.
2. A convex function on an open convex set is always continuous.
3. If f is convex on $[0, 1]$, then f must be continuous at $x = 0.5$.
4. A convex function can have a jump discontinuity in the interior of its domain.

Hint: What does the theorem say about interior vs boundary?

Solution: Continuity of Convex Functions

Answers

1. “Every convex function on $\mathbb{R}^n$ is continuous everywhere”: **TRUE**

Since $\mathbb{R}^n$ is open, $\text{int}(\mathbb{R}^n) = \mathbb{R}^n$, so continuity holds everywhere.

2. “Convex on open convex set $\Rightarrow$ continuous”: **TRUE**

An open set equals its interior, so the theorem applies everywhere.

3. “ f convex on $[0, 1] \Rightarrow$ continuous at 0.5”: **TRUE**

$x = 0.5$ is in the interior of $[0, 1]$, so continuity is guaranteed.

4. “Jump discontinuity possible in interior”: **FALSE**

The theorem guarantees continuity in the interior. Discontinuities can only occur at boundary points.

Roadmap

Convex and Concave Functions

Strictly Convex and Strictly Concave Functions

Examples of (Strictly) Convex and Concave Functions

Characterizations of Convex Functions

Proving Convexity via Composition Rules

Calculus Background

First-Order Condition for Convexity

Second-Order Calculus Background

Second-Order Condition for Convexity

Operations Preserving Convexity of Functions

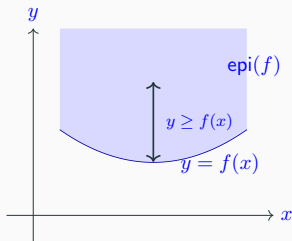
Continuity of Convex Functions

Epigraph and Hypograph

Epigraph and Hypograph of a Function

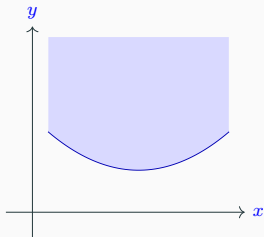
Let S be a nonempty set in $\mathbb{R}^n$ and let $f : S \rightarrow \mathbb{R}$.

- **Graph:** $\text{gph}(f) := \{(\mathbf{x}, f(\mathbf{x})) : \mathbf{x} \in S\} \subset \mathbb{R}^{n+1}$.
- **Epigraph:** $\text{epi}(f) := \{(\mathbf{x}, y) : \mathbf{x} \in S, y \in \mathbb{R}, y \geq f(\mathbf{x})\} \subset \mathbb{R}^{n+1}$.
- **Hypograph:** $\text{hyp}(f) := \{(\mathbf{x}, y) : \mathbf{x} \in S, y \in \mathbb{R}, y \leq f(\mathbf{x})\} \subset \mathbb{R}^{n+1}$.

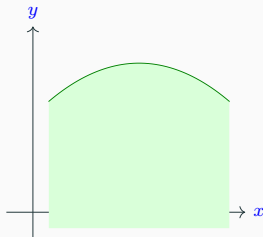


The **epigraph** is everything “above” the graph; the hypograph is everything “below.”

Epigraph vs Hypograph



Epigraph of Convex f
 $\text{epi}(f)$ is CONVEX



Hypograph of Concave g
 $\text{hyp}(g)$ is CONVEX

Key Observations

- Epigraph of a convex function is a convex set
- Hypograph of a concave function is a convex set
- This connects function convexity to set convexity!

Epigraph: Link Between Convex Sets and Functions

Theorem

Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. Then f is a convex function if and only if $\text{epi}(f)$ is a convex set.

Proof Sketch:

($\Rightarrow$) Assume f is convex. Take $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2) \in \text{epi}(f)$.

Then $y_1 \geq f(\mathbf{x}_1)$ and $y_2 \geq f(\mathbf{x}_2)$.

For $\lambda \in [0, 1]$:

$$\begin{aligned}\lambda y_1 + (1 - \lambda)y_2 &\geq \lambda f(\mathbf{x}_1) + (1 - \lambda)f(\mathbf{x}_2) \\ &\geq f(\lambda \mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2) \quad (\text{convexity of } f)\end{aligned}$$

So $(\lambda \mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2, \lambda y_1 + (1 - \lambda)y_2) \in \text{epi}(f)$. ✓

($\Leftarrow$) Exercise (use similar argument in reverse).

Consequences of the Epigraph Characterization

Corollary 1

The hypograph of a concave function is a convex set.

Proof: If f is concave, then $-f$ is convex, so $\text{epi}(-f)$ is convex. But $\text{hyp}(f) = \{(\mathbf{x}, y) : y \leq f(\mathbf{x})\} = \{(\mathbf{x}, -z) : z \geq -f(\mathbf{x})\}$, which is a reflection of $\text{epi}(-f)$.

Corollary 2

Since the epigraph of a convex function is a convex set, it has supporting hyperplanes at boundary points.

This leads to **subgradients** (coming in Lecture 5)!

Why This Matters

- Reduces function problems to set problems
- All theorems about convex sets apply to epigraphs
- Foundation for duality theory: separating hyperplanes for epigraphs

Exercise: Working with Epigraphs

Epigraph Exploration

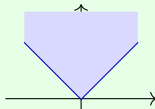
1. Sketch the epigraph of $f(x) = |x|$ on $\mathbb{R}$. Is it a convex set?
2. Let $f(x) = x^2$ and $g(x) = 2x + 1$. Describe $\text{epi}(f) \cap \text{epi}(g)$. What function does this represent the epigraph of?
3. If $\text{epi}(f)$ is a polyhedron (intersection of finitely many halfspaces), what can you conclude about f ?

Hint: For (2), think about $\max(f, g)$.

Solution: Working with Epigraphs

Answers

1. **Epigraph of $|x|$:** The region above the V-shape. Yes, it's convex (the intersection of two halfplanes: $y \geq x$ and $y \geq -x$).



2. **$\text{epi}(f) \cap \text{epi}(g)$:** This is $\text{epi}(\max(f, g)) = \text{epi}(\max(x^2, 2x + 1))$.
The intersection of epigraphs gives the epigraph of the pointwise maximum.
3. **Polyhedral epigraph:** f must be a **piecewise linear convex function** (maximum of finitely many affine functions).

Quick Review (5 questions)

Answer True or False:

1. The function $f(x) = e^{-x^2}$ is convex on $\mathbb{R}$.
2. If f and g are both convex, then $f - g$ is convex.
3. A strictly convex function can have at most one global minimizer.
4. The epigraph of any function is always a convex set.
5. If f is convex on a closed interval $[a, b]$, then f is continuous on (a, b) .

End of Lecture Assessment: Solutions

Answers

1. $f(x) = e^{-x^2}$ is convex: **FALSE**

$-x^2$ is concave; e^t is convex and increasing. Composition of convex increasing with concave gives neither in general. In fact, e^{-x^2} is the Gaussian bell curve—it's neither convex nor concave globally.

2. $f - g$ convex if f, g convex: **FALSE**

g convex $\Rightarrow -g$ concave. Sum of convex and concave is generally neither.

3. Strictly convex has at most one minimizer: **TRUE**

Two distinct minimizers would contradict strict convexity (the midpoint would have strictly smaller value).

4. Epigraph always convex: **FALSE**

Only true if f is convex. Epigraph of $-x^2$ is not convex.

5. f convex on $[a, b] \Rightarrow$ continuous on (a, b) : **TRUE**

(a, b) is the interior of $[a, b]$, and convex functions are continuous on the interior.

Summary and Next Steps

Key Takeaways

- **Definition:** $f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) \leq \lambda f(\mathbf{x}_1) + (1 - \lambda) f(\mathbf{x}_2)$
- **First-order condition:** $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x})$
- **Second-order condition:** $\nabla^2 f(\mathbf{x}) \succeq 0$
- **Preservation rules:** sums, max, affine composition, monotone composition
- **Epigraph:** f is convex $\Leftrightarrow \text{epi}(f)$ is a convex set

Next Time (Lecture 5):

- Subgradients for non-differentiable convex functions
- Optimality conditions: $\mathbf{0} \in \partial f(\bar{\mathbf{x}})$
- Subdifferential calculus

Questions?